

Challenge (Habanero)

- Q1.** What type of correlation is shown by the data points (2, 4), (4, 6), (6, 8), and (8, 10) in a scatter plot?
- (A) Negative
(B) Positive
(C) None
(D) Inverse
(E) Direct
- Q2.** A hiker tracks her distance d (in miles) and time t (in hours) to reach the top of a mountain. If the data points are (2, 4), (3, 5), (5, 7), what is the equation of the line of best fit?
- (A) $d = 2t - 1$
(B) $d = t + 1$
(C) $d = 2t + 1$
(D) $d = t - 2$
(E) $d = 3t - 2$
- Q3.** The fuel consumption y (in gallons) of a car is measured for different distances x (in miles) traveled. If the data points are (10, 2), (20, 4), (30, 6), what is the correlation between x and y ?
- (A) Negative
(B) Positive
(C) None
(D) Inverse
(E) Direct
- Q4.** A park ranger records the population P of deer and the amount of food F (in pounds) available in the park. If the data points are (10, 20), (15, 30), (20, 40), what is the equation of the line of best fit?
- (A) $P = 2F - 10$
(B) $P = F + 5$
(C) $P = 2F + 5$
(D) $P = F - 10$
(E) $P = 3F - 20$
- Q5.** A scatter plot shows the relationship between the number of hours h a hiker spends walking and the distance d (in miles) covered. If the correlation is positive, which of the following statements is true?
- (A) As h decreases, d increases
(B) As h increases, d decreases
(C) As h increases, d increases
(D) As h decreases, d remains constant
(E) As h increases, d remains constant
- Q6.** The data points (1, 3), (2, 5), (3, 7) represent the number of days x a group of friends spend hiking and the total distance y (in miles) they cover. What is the predicted value of y when $x = 4$?
- (A) 6
(B) 7
(C) 8
(D) 9
(E) 11
- Q7.** A study shows the relationship between the amount of rainfall R (in inches) and the growth G (in inches) of a certain plant species. If the data points are (2, 4), (4, 6), (6, 8), what is the rate of change of G with respect to R ?
- (A) 1 inch per inch
(B) 2 inches per inch
(C) 3 inches per inch
(D) 4 inches per inch
(E) 0.5 inches per inch
- Q8.** The cost C (in dollars) of a hiking trip is related to the number of participants p . If the data points are (2, 100), (4, 200), (6, 300), what is the equation of the line of best fit?
- (A) $C = 50p$
(B) $C = 50p + 50$
(C) $C = 50p - 50$
(D) $C = 25p$
(E) $C = 25p + 25$
- Q5.** A scatter plot shows the relationship between the number

Open Ended Questions (FRQ)

- Q9.** A group of friends go on a hiking trip and record their distance d (in miles) and time t (in hours) to reach the top of a mountain. The data points are (2, 4), (4, 6), (6, 8). Draw a scatter plot and find the equation of the line of best fit. Use the equation to predict the time it will take to cover a distance of 10 miles.
- Q10.** A park ranger records the population P of deer and the amount of food F (in pounds) available in the park. The data points are (10, 20), (15, 30), (20, 40). Create a scatter plot and determine the correlation between P and F . Use the data to predict the population of deer when there are 35 pounds of food available.

Chef's Tips

- Tip 1: Ensure students use the correct units for their calculations.
- Tip 2: Encourage students to label their axes correctly in the scatter plots.
- Tip 3: Remind students to check their equations of the line of best fit for reasonableness.